

Classification Algorithms for Web User Identification

ST – O13: Statistical Learning and Data Mining – Assignment 1
Sep-Oct 2019

Pushkar Kale (1825)
Mrunmayi Naik (1836)

Table of Contents

PROBLEM STATEMENT	3
OBJECTIVES	3
DATA DESCRIPTION AND FEATURE ENGINEERING	4
EXPLORATORY ANALYSIS	6
IMPLEMENTATION OF THE PROGRAM	10
CLASSIFICATION ALGORITHMS	11
RESULTS AND DISCUSSION	15
SELECTING BEST MODEL	16
PREDICTIONS	18
CONCLUDING REMARKS	19
REFERENCES	20

Problem Statement

Web-user identification is one of the important research topics in the behavioral psychology. Here we try to identify a user on the Internet by tracking his/her sequence of visited Web pages. The algorithm to be built will take a webpage session (a sequence of ten webpages visited consecutively by the same person) along with the time stamps of visit for a webpage and predict whether it belongs to Alice or an intruder.

Objectives

- Try to extract some useful information from the train data by creating some new variables using the given variables
- To build a classifier using the train data
- To select the best model using 10-fold CV error
- To determine the usefulness of the variables with visualizations
- Prediction on the test data

Data Description and Feature Engineering

The given data is about the behavior of Alice and an intruder over about 7000 web sessions. It considers the duration from 12-03-2013 to 30-04-2014. Each webpage session is a sequence of ten webpages visited consecutively by the same person (Alice or an intruder).

The original variables are - Site ids of each website visited in each session along with its time and date labelled as 1 or 0 (Alice or an intruder).

The Modified data consist of 26 variables (including dummy variables). Following variables are created using the existing variables.

Modified using Excel:

1. Time Spent in a Session (Time Spent in Mins): The difference between visit time of site10 and site1 in minutes is computed.
2. Start Hour of a session (Start Hour): Start Hour is recorded as the hour when first site in a session is viewed, that is when the session starts.
3. Day of the Month (Day): Date of the Month for each session is extracted from the Time Stamp.
4. Month of the Year (Month): Month for each session is extracted from the Time Stamp.

Modified using R:

1. Number of sites visited in a session (No_of_sites): The total number of unique sites visited in each session.
2. Each site is classified into three non-overlapping groups:
 - i) Unique sites visited by the intruder
 - ii) Unique sites visited by Alice
 - iii) Sites common to both

The above groups are named as say for Site 1: Set1.0, Set1.1 and Set1.2 respectively.

Two dummy (3-1) variables created for the groups i.e. total 20 dummy variables.

Illustration:

Suppose the three non-overlapping groups are:

Group 1: 729, 856, 1825, 7297, 8997, 21

Group 2: 345, 4567, 2390, 11456

Group 3: 29, 1490, 4323, 6789

Table 1 is the original data and Table 2 is the transformed data. Note that, only 2 indicator variables are required for 3 levels.

Session	Site1
1	856
2	11456
3	1825
4	29
5	21

Table 1

Session	Set1.0	Set1.1
1	1	0
2	0	1
3	1	0
4	0	0
5	1	0

Table 2

Following is the list of all the variables in Modified Train Data.

```
'data.frame': 6958 obs. of 26 variables:
 $ Set1.0      : num 1 0 0 1 1 0 1 1 1 1 ...
 $ Set1.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set2.0      : num 1 1 0 1 0 0 1 1 0 1 ...
 $ Set2.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set3.0      : num 0 1 0 0 1 0 1 1 0 0 ...
 $ Set3.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set4.0      : num 1 0 0 0 1 0 0 1 0 0 ...
 $ Set4.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set5.0      : num 1 1 0 0 1 0 1 1 1 0 ...
 $ Set5.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set6.0      : num 1 0 0 0 1 0 0 1 0 1 ...
 $ Set6.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set7.0      : num 1 1 0 0 1 0 1 1 1 1 ...
 $ Set7.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set8.0      : num 1 0 0 0 1 0 1 0 1 1 ...
 $ Set8.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set9.0      : num 1 0 0 1 1 0 1 0 0 0 ...
 $ Set9.1      : num 0 0 0 0 0 0 0 0 0 0 ...
 $ Set10.0     : num 1 0 0 1 1 0 1 1 1 0 ...
 $ Set10.1     : num 0 0 0 0 0 0 0 0 0 0 ...
 $ No_of_sites : int 8 9 2 10 8 2 9 5 4 4 ...
 $ Start.Hour  : int 11 14 17 17 18 8 8 8 8 8 ...
 $ Time.Spent.In.Mins.: int 9 3 1 0 2 5 4 0 1 0 ...
 $ target      : int 0 0 0 0 0 0 0 0 0 0 ...
 $ Day         : int 12 12 12 12 12 12 12 12 12 12 ...
 $ Month       : int 1 1 1 1 1 2 2 2 2 2 ...
```

Figure 1

Exploratory Analysis

1) Comparison of Start Hour for both the users

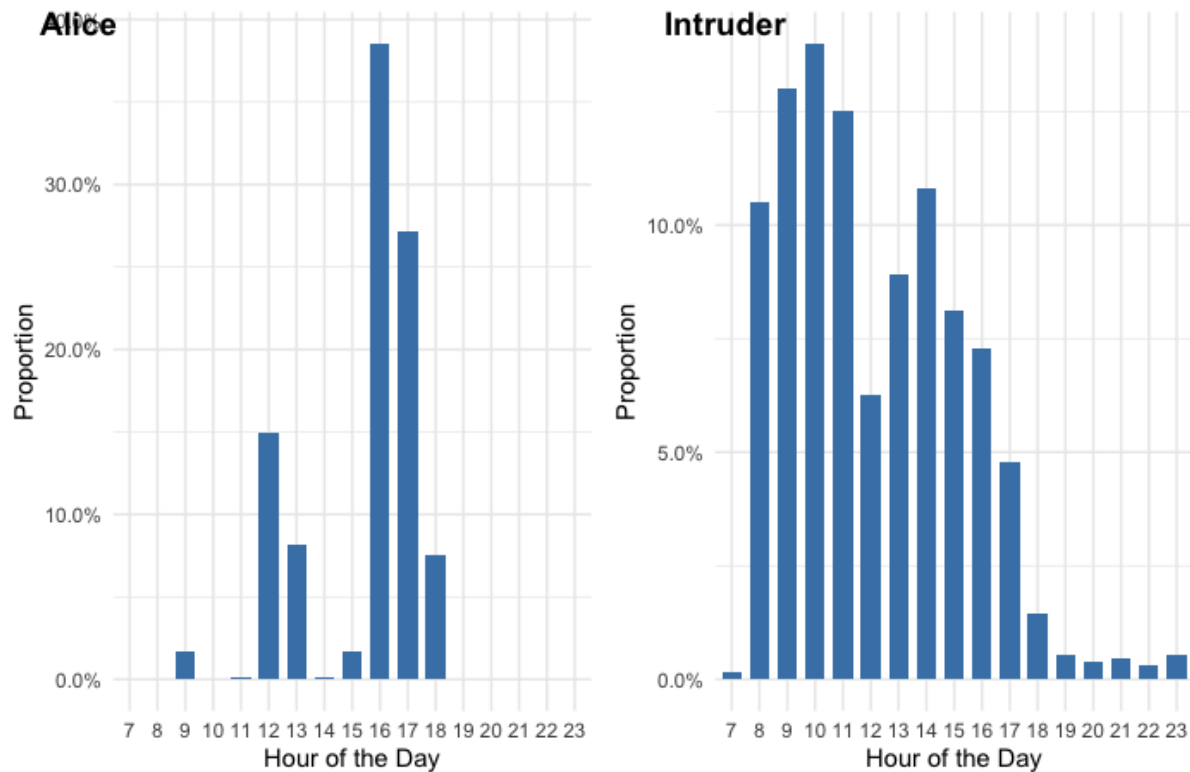


Figure 2

Generally, different people prefer to visit webpages at different times of the day. From this graph, it is clear that Alice and Intruder's time of visits have different patterns. Majority of Alice's visits are in between 4 to 6 p.m. with some of the visits are during early hours of afternoon. Intruder has visits starting from morning hours to late night. We can say that a session that is started between 8 to 11 a.m. is more likely to belong to an intruder than Alice.

2) Number of unique sites visited in a session

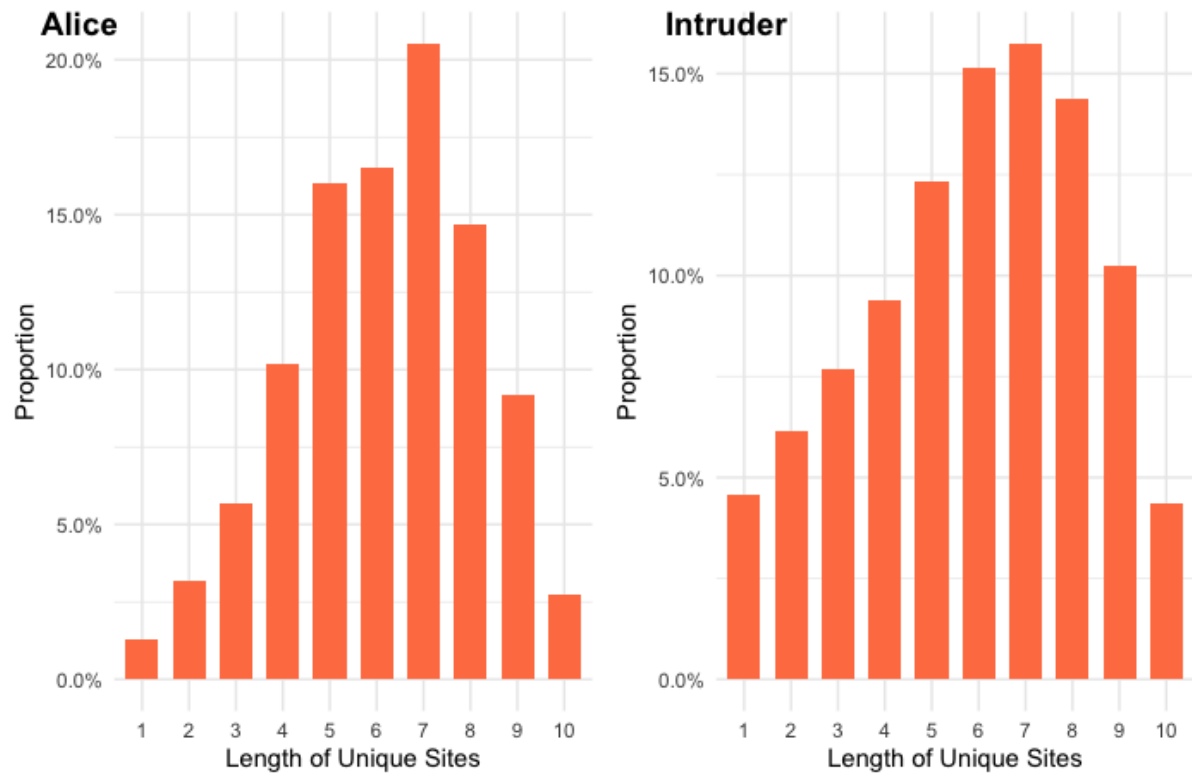


Figure 3

Many times, we visit a webpage multiple times in a single session of 10 webpages. This behaviour is reflected in these bar plots. Intruder is more likely to visit a same website in a single session than Alice. Otherwise, there is not much difference in the pattern of both the plots.

3) Time Spent in a Session

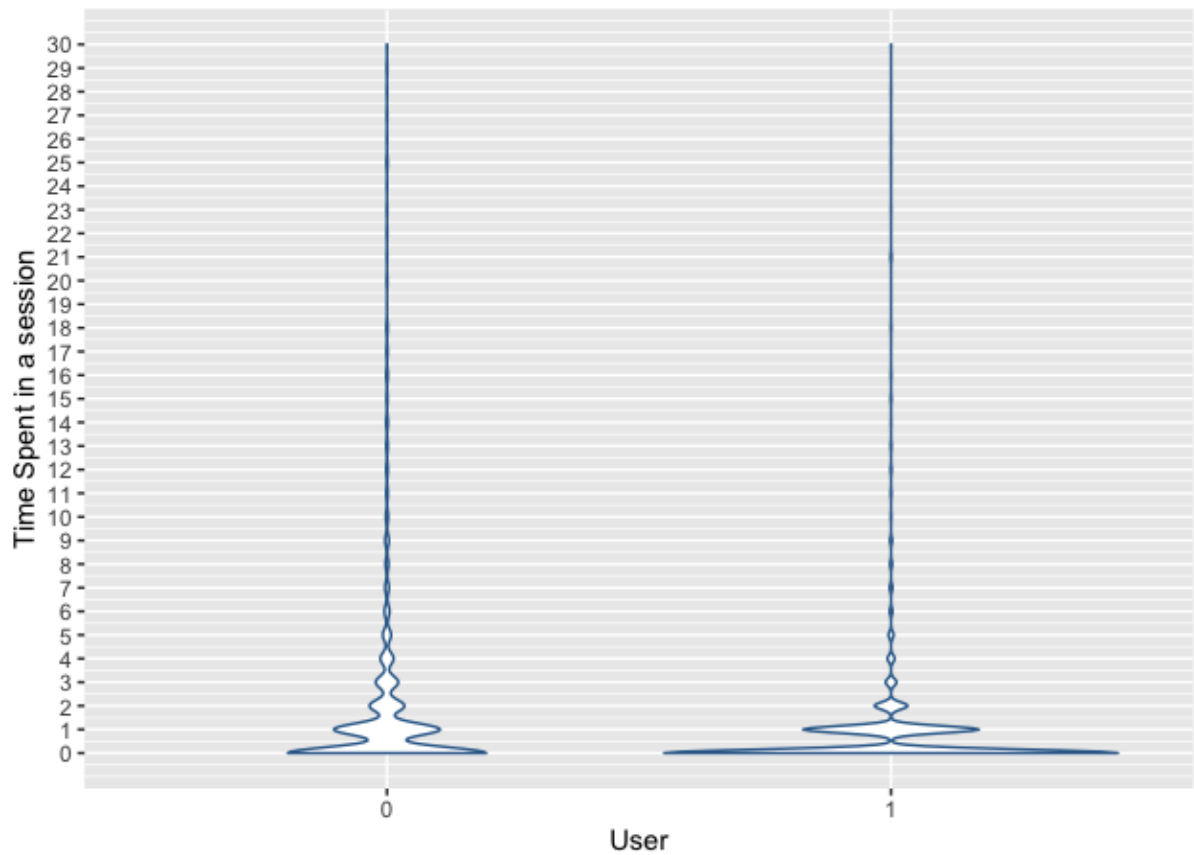


Figure 4

This violin plot is showing the difference between the distribution of duration of a session by user. Alice's sessions mostly have very short spans (less than 1 minute) while for intruder, there are significant number of instances with duration more than 2 minutes.

4) Site wise behaviour of the Users

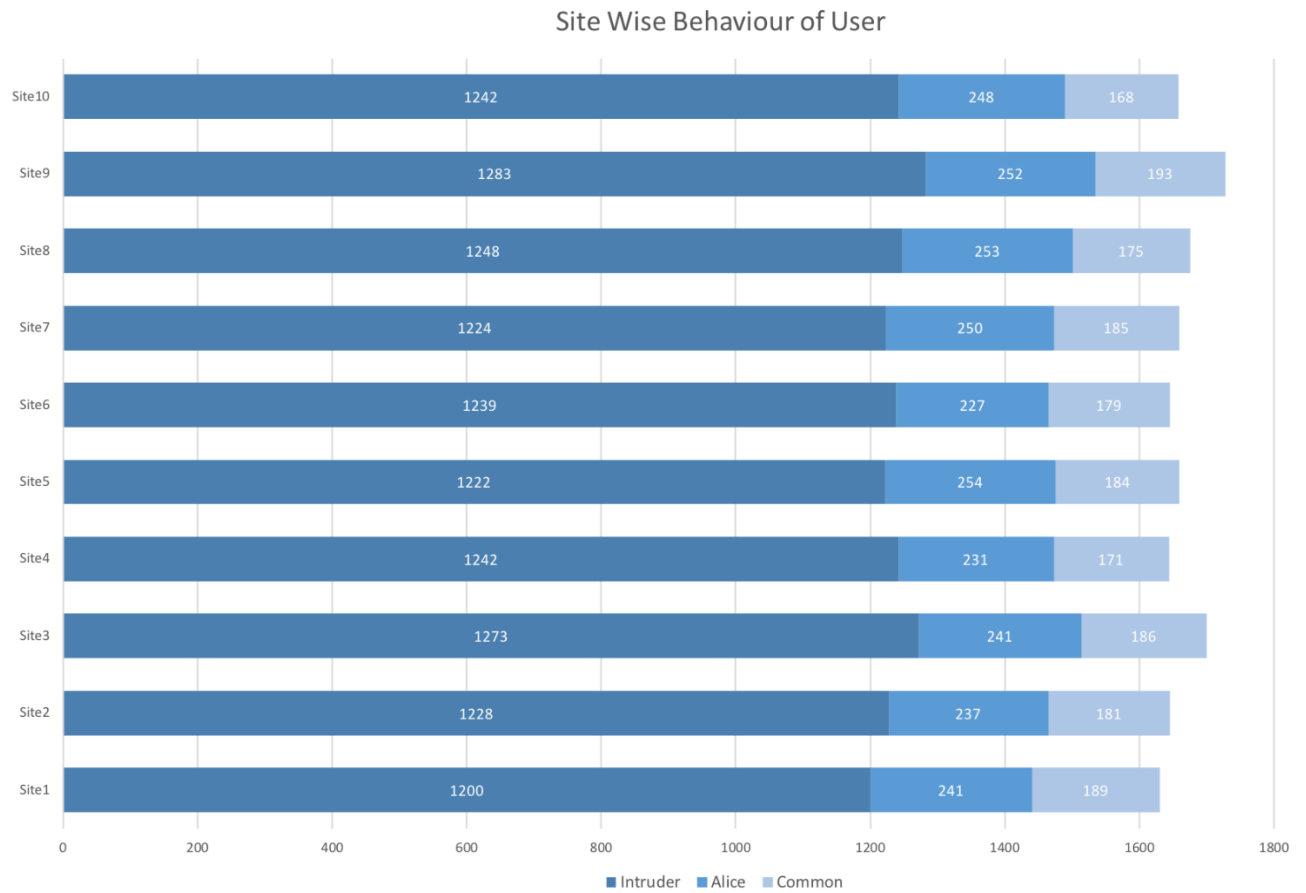


Figure 5

For each site in a session, there are about 170-200 common webpages for Alice and an intruder. Though, we have more data points (Almost twice) of an intruder than Alice, number of unique sites is still far larger than (5 times) for an intruder than Alice.

Implementation of the Program

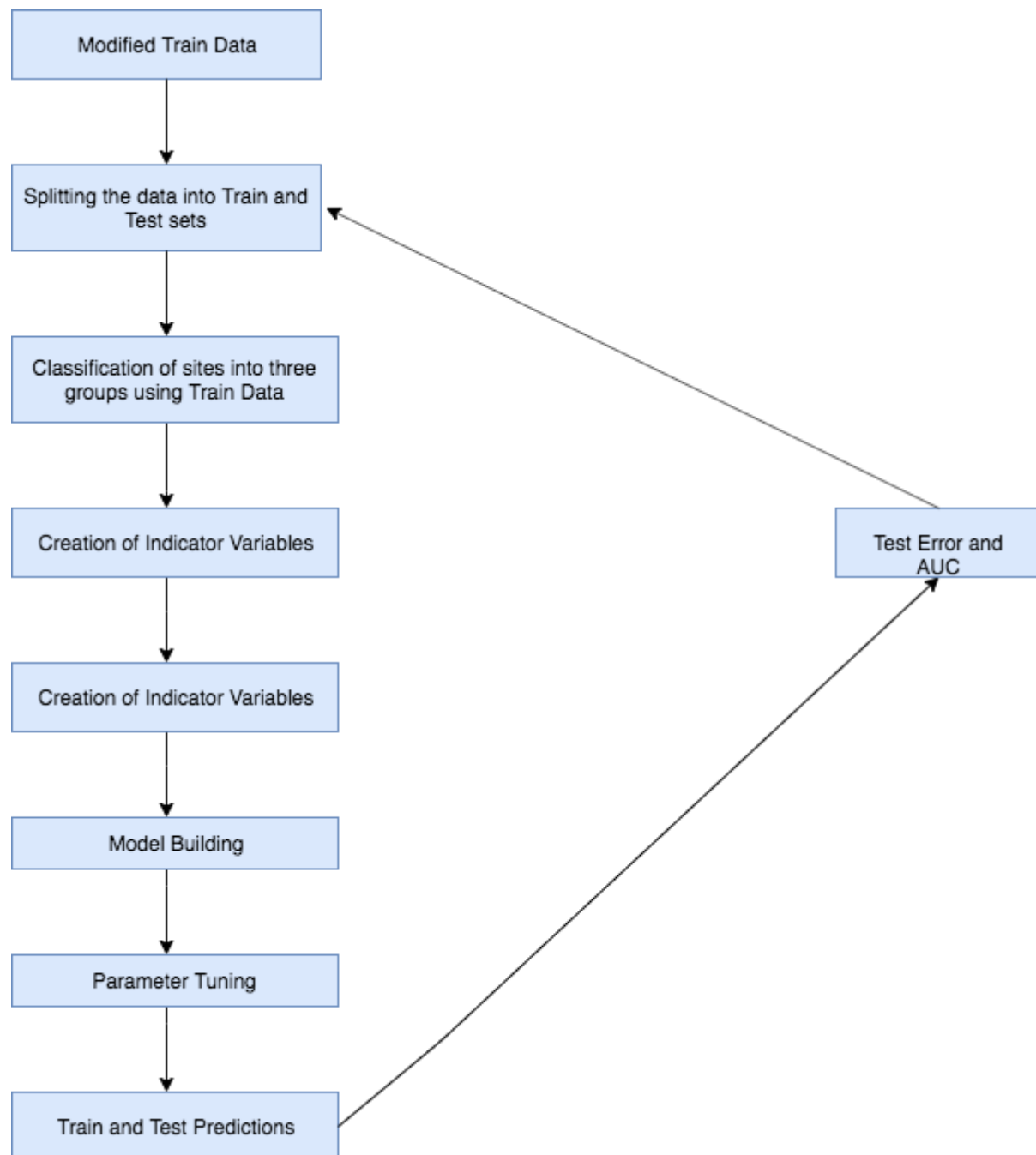


Figure 6

Above loop would run 10 times from which we can get 10-fold CV error and 10-fold AUCs and Coeff. of Variation (for Error) for the each of the classifier. The entire R-code for this algorithm can be found in [Data_Classifiers.R](#).

Classification Algorithms

1) Classification Tree

A classification tree makes use of recursive binary partitions to obtain predictions. Interpretation of a tree is very easy, hence it is used to identify important variables and observe their role in classification.

A tree fit the Train Data is as follows :

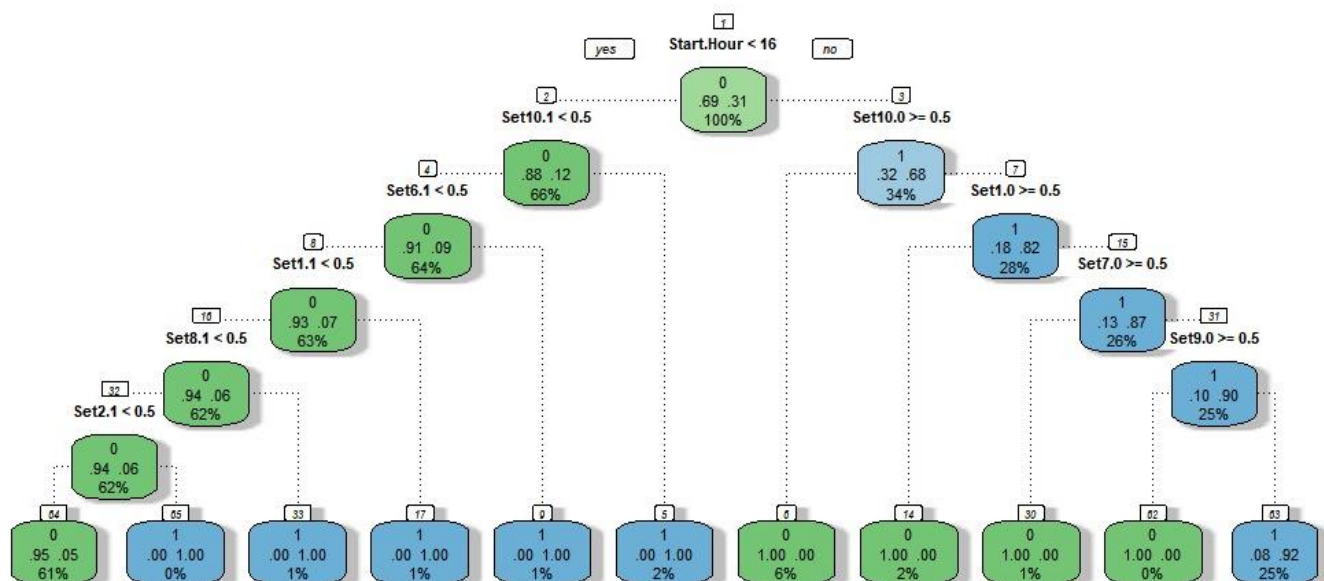


Figure 7

The first split is based on the variable Start Hour. The hour at which the session starts gives us an idea about the identity of the user, as Alice seems to prefer the time later than 4 pm.

The subsequent splits are mostly based on the sites viewed by the user. In the rightmost branches, we observe that the modal prediction is 1 for nodes with indicator variable $\text{SetA.0} = 0$ ('No' branches for ' $\text{SetA.0} \geq 0.5$ ' in the tree, A is any one of the sites 1 to 10).

This can be easily interpreted, as the site viewed by the user is the one that Alice has already viewed, the probability that the user is Alice is higher than her being an intruder.

Similarly the left most branches show modal prediction is 0 for nodes with indicator variable $\text{SetA.1} = 0$ ('Yes' branches for ' $\text{SetA.1} < 0.5$ ' in the tree, A is any one of the sites 1 to 10).

This is interpreted as, the site viewed by the user is the one that the intruder has viewed before, the probability that the user is an intruder is higher than her being Alice.

2) Random Forest

Random Forest gives better prediction accuracy than trees by giving a more robust model. The model consists an aggregate of a number of trees fitted on the data, considering 20 randomly selected variables each time. The parameter number of variables (mtry) value was decided using tuning.

Random Forest helps in identification of important variables in classification by ranking them according to the mean decrease in Gini index. The Variable Importance plot is as follows:

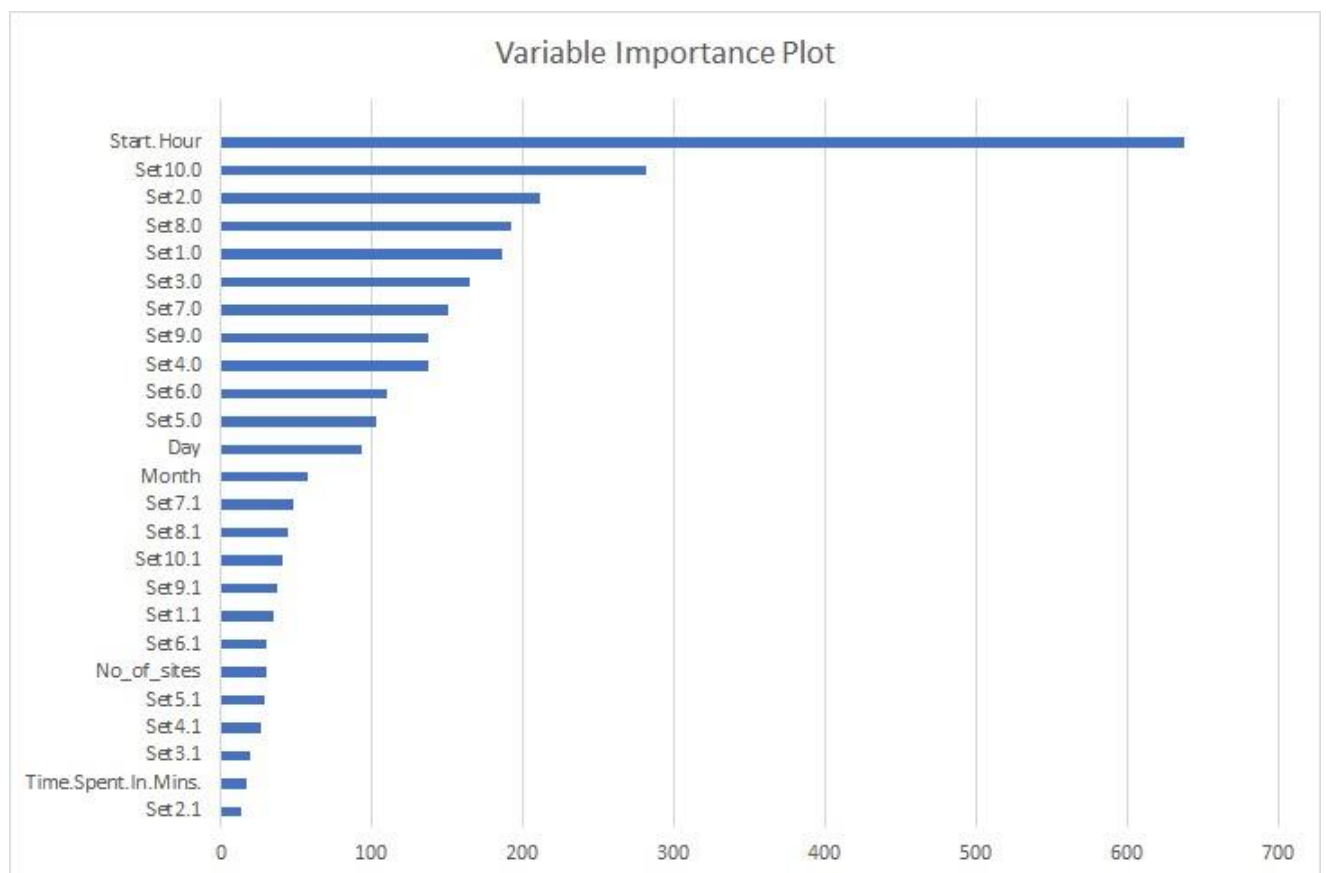


Figure 8

With maximum mean decrease in Gini index, the variable 'Start Hour' is the most important variable in identifying the user (this seems consistent with the figure 3, which is indicating clear difference in two users). It is followed by the indicator variables for Site 2 and Site 10 which are known to be viewed by an intruder. The subsequent variables are for the rest of the site groups, those common with the intruder having more importance.

3) Boosting

Another attempt to increase the prediction accuracy of classification tree is Boosting. It is an ensemble method for improving the model predictions, by training weak learning algorithms sequentially. A generalised boosted model is fit on the data. The variables 'Start Hour' and Set10.0 (an indicator variable for whether the last site visited in a session matches with intruder) are turned out to be most important. Model parameters are tuned using 'caret' package in R. Following are the values of parameters :

n.trees: Total no. of trees to fit = 150

interaction.depth: The highest level of variable interaction allowed = 3

shrinkage: Learning rate or step-size reduction = 0.1

n.minobsinnode: Minimum no. of obs. in the terminal nodes = 10

The 10-fold CV error is lower than Random Forest.

4) Naïve Bayes Classifier

A Naïve Bayes Classifier is a probabilistic machine learning model that is used for classification task. The crux of the classifier is based on Bayes theorem. This algorithm is more efficient in terms of time taken and memory, but its 10-fold CV error is highest among the all the classifiers that we have tried. A-priori probabilities are taken as 0.6896 for '0' and 0.3104 for '1' i.e. same as proportions in the train data. We discard this classifier on the basis of predictive accuracy, which is not satisfactory.

5) Logistic Regression with Bayesian approach*

It is the go-to method for binary classification problems. Logistic regression uses an equation as the representation, very much like linear regression. For the analysis, we have used a Bayesian approach instead of Classical approach. The main advantage of this method is that MLEs of the parameters of a Non Bayesian Regression model overfits the data, meaning the unknown value for a certain value of independent variable becomes too precise when calculated. Bayesian Linear Regression relaxes this fact, saying that there is uncertainty involved by incorporating 'Predictive Distribution'. Prior distribution is selected as default i.e. Cauchy with mean 0 and scale 2.5.

Almost all the variables are turned out as significant with the p-value of 'Start Hour' is almost zero. The predictive accuracy of this model is also really good with 10-fold CV error of only about 10%.

6) Support Vector Machines

A Support Vector Machine (SVM) is a discriminative classifier formally defined by a separating hyperplane. In other words, given labelled training data (supervised learning), the algorithm outputs an optimal hyperplane which categorizes new examples. In two dimensional space this hyperplane is a line dividing a plane in two parts where in each class lay in either side.

Here, we have used `svm()` function from 'e1071' package. After tuning, following values of parameters are selected.

Cost: cost of constraints violation = 4

Gamma: Parameter required for Kernel = 0.1

After trying 2-3 different kernels, we have selected 'radial' kernel based on accuracy.

Here is the summary of the model:

```
Parameters:
  SVM-Type:  C-classification
  SVM-Kernel: radial
           cost: 4

Number of Support Vectors: 1411

( 860 551 )

Number of Classes: 2

Levels:
0 1
```

Figure 9

In terms of performance on test data, this model is the best among all tried models. (See Table 3)

Results and Discussion

The 10 – fold cross validation over all the models was performed. The models are compared using the following:

- i) Train Error
- ii) Test Error
- iii) Area under the ROC Curve
- iv) Coefficient of Variation for Test Accuracies

The results are as follows:

Model	Train Accuracy	Test Accuracy	AUROC	Coeff. of Variation
Naïve Bayes Classifier	0.942368	0.817762	0.796820	0.022359
Logistic Regression	0.972502	0.884593	0.861812	0.014819
Support Vector Machine	0.995609	0.938200	0.924341	0.009953
Classification Tree	0.956996	0.875680	0.850381	0.017349
Random Forest	0.989860	0.899396	0.886216	0.011407
Boosting	0.982386	0.920667	0.903525	0.009809

Table 3

All the models have high train accuracy. In fact, for SVM, train error is less than 1%. But sometimes train accuracy can be misleading as high values of those may be a result of overfitting. 10-fold CV Error is a good indicator of predictive performance of the model. Here as well SVM is performing quite well (With only 6.18% error).

Classification Tree, Random Forest and Logistic Regression algorithms have similar performance on the test data (About 10-12% error). Boosting Model is also performing very well on the test data with about 8% CV error. Though Naïve Bayes has the poorest accuracy among all, it has 19% error, which is not bad.

Area Under ROC Curve (AUROC) is one of the most important evaluation metrics for checking any classification model's performance. The ROC curve is plotted with True Positive Rates (TPR) against the False Positive Rates (FPR) where TPR is on y-axis and FPR is on the x-axis.

SVM performs the best among all in this aspect as well. AUC of 0.9382 means there is about 94% chance that model will be able to differentiate between positive and negative class. Boosting has performance almost as good as SVM. All other models except Naïve Bayes have AUROC between 0.8 to 0.9, which is also reasonably good.

From coefficient. of variation, it is evident that stability of SVM model along with Boosting is very good.

Selecting Best Model

From Table 3, we have two models competing for the ‘best’ model. Both SVM and Boosting have performing exceptionally well with 10-fold CV accuracy greater than 90% with almost equal coefficients of variation. Also, AUROC for both the models is more than 0.9.

Hence, we need to have some different measures for selecting the best model.

Confusion Matrix and Statistics

```
pred_boost_test
  0  1
0 448 34
1  23 191
```

Accuracy : 0.9181
95% CI : (0.8952, 0.9374)
No Information Rate : 0.6767
P-Value [Acc > NIR] : <2e-16

Kappa : 0.8104

Mcnemar's Test P-Value : 0.1853

Sensitivity : 0.9512
Specificity : 0.8489
Pos Pred Value : 0.9295
Neg Pred Value : 0.8925

Boosting

Confusion Matrix and Statistics

```
Pred
  0  1
0 453 29
1  22 192
```

Accuracy : 0.9267
95% CI : (0.9048, 0.945)
No Information Rate : 0.6825
P-Value [Acc > NIR] : <2e-16

Kappa : 0.8295

Mcnemar's Test P-Value : 0.4008

Sensitivity : 0.9537
Specificity : 0.8688
Pos Pred Value : 0.9398
Neg Pred Value : 0.8972

SVM

Figure 10

From the confusion matrices of both the models, there is not much difference between the both the models. If we see the 95% C.I. for test accuracy, lower limit for SVM is also above 90. Also, Specificity of SVM model is better than Boosting. Also, if we see Misclassification table, Boosting model is misclassifying 0 as 1 more times than SVM. In our problem, classifying an intruder as Alice is more serious error than classifying Alice as an intruder.

Hence, we select SVM as the best model and will make final predictions using the same.

Note: All these computations are done by fitting model on 90% data (Train) and predicting for remaining 10% data (Test).

ROC Curve for SVM:

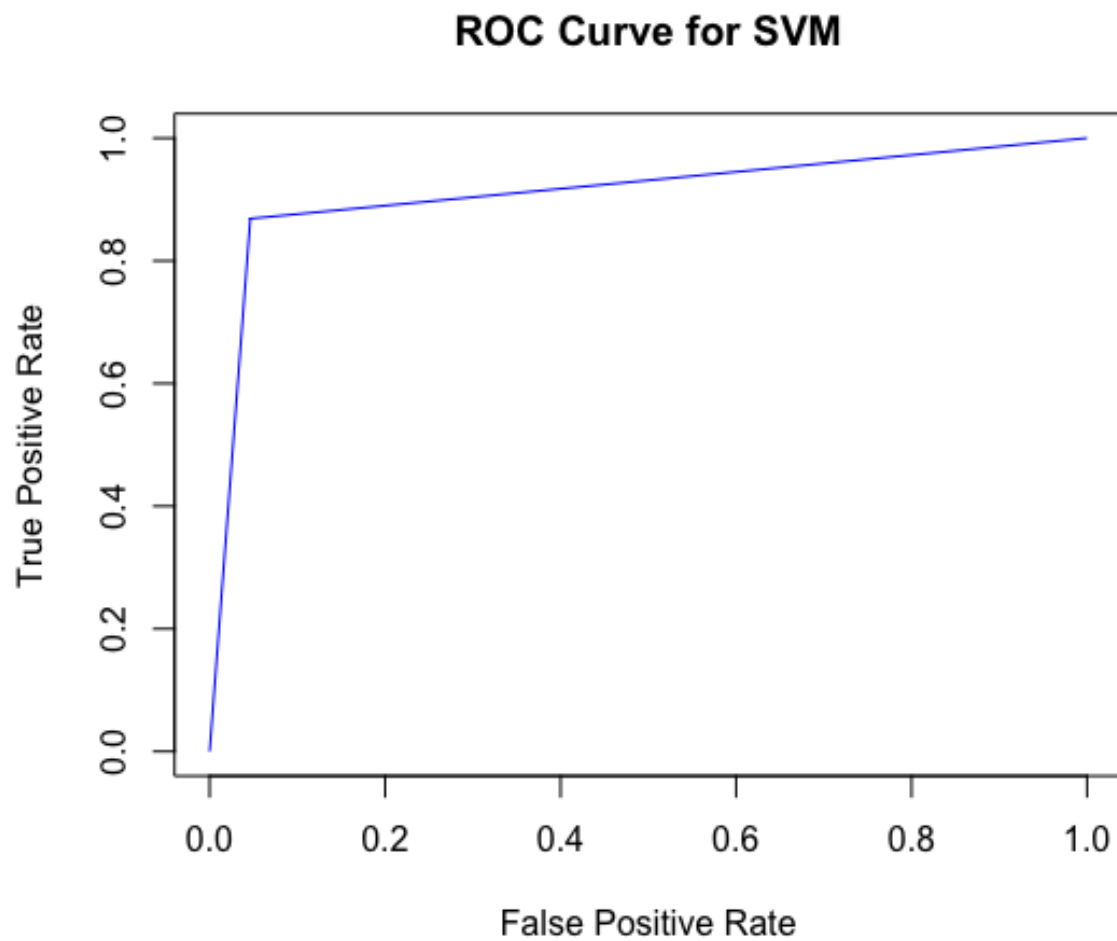


Figure 11

As evident from AUROC, ROC curve for SVM is exceptionally well. It is very close to an ideal ROC curve.

Predictions

A Test Data have 300 observations for which target is unknown. We need to predict the target values (0 or 1).

For creation of indicator variables for site groups, all the given train data is considered i.e. for test data, all the indicator variables are created using Train Data only. All other variables (Start Hour, Day, Time spent etc.) are created by using same procedure and software as done for Train Data.

The number of 0's predicted is 198, while that of 1's predicted is 102. This seems to be consistent with the proportion in the Train Data.

Concluding Remarks

The given data was modified by creating new variables from the existing ones. Six different classifiers were fit on this modified data. These classifiers were compared on the basis of 10-fold cross validation error and area under the ROC. SVM has the highest AUROC and is hence considered the best among all and used for prediction.

SVM with radial kernel, gamma parameter as 0.1 and cost equal to 4 is the final classifier chosen. The 10-fold cross validation error for this model is 6.18%.

The predictions for the test data are to be found in file :

[“Predictions\(Assignment 1\) - 1825,1836.csv”](#).

The variable 'Start Hour' turns out to be the most important variable. Exploratory analysis (Fig 2) gives us an idea about the difference in behaviour of the users in terms of the time session starts. This is further evident in the Variable Importance Plot (Fig 8) in Random Forest, it contributes to the maximum mean decrease in Gini index and hence is the most important variable in identifying the user.

'Set 10.0' which records whether the 10th webpage visited by the user is the one previously visited by the intruder is the next important variable, as observed in the Variable Importance Plot. These two variables are the first two splits in the tree fitted on the data (Fig 7). Thus, time of access and the webpages viewed at the end can be regarded as the most important variables in classification.

The drawback of using SVM is that interpretability of model is lost. Though the prediction accuracy is very good, we cannot infer the role played by the variables easily. Tuning the model for better results is time consuming.

We have considered only a few basic models for this analysis. Exploring advanced models like neural networks and deep learning techniques will certainly help improve the prediction accuracy.

We hope that the performance of our classifier will improve with increase in the number of train data points.

References

- 1) Robert Tibshirani, Daniel Witten, Gareth James, Trevor Hastie. An Introduction to Statistical Learning with Applications in R. Springer. (2013)
- 2) Trevor Hastie, Robert Tibshirani, Jerome Friedman. The Elements of Statistical Learning: Data Mining, Inference and Predictions. Springer. (2008)
- 3) Towards Data Science: Sharing concepts, ideas and codes <https://towardsdatascience.com/>
- 4) Medium: Data Science. <https://medium.com/topic/data-science>
- 5) DataCamp. <https://www.datacamp.com/>
- 6) SuperDataScience. <https://www.superdatascience.com/blogs>